BreakBite

Pre-order your food in the break between two classes, collect it at the counter.

PRM393 · Lab 3 · Cloud-Enabled Mobile App (Firebase, ARTEMIS, AI code review)

Team BreakBite · Campus District 9 · Flutter + Firebase

Repository: https://github.com/baonv0817/PRM393_Lab03_Group2 · deliverables map in README.md

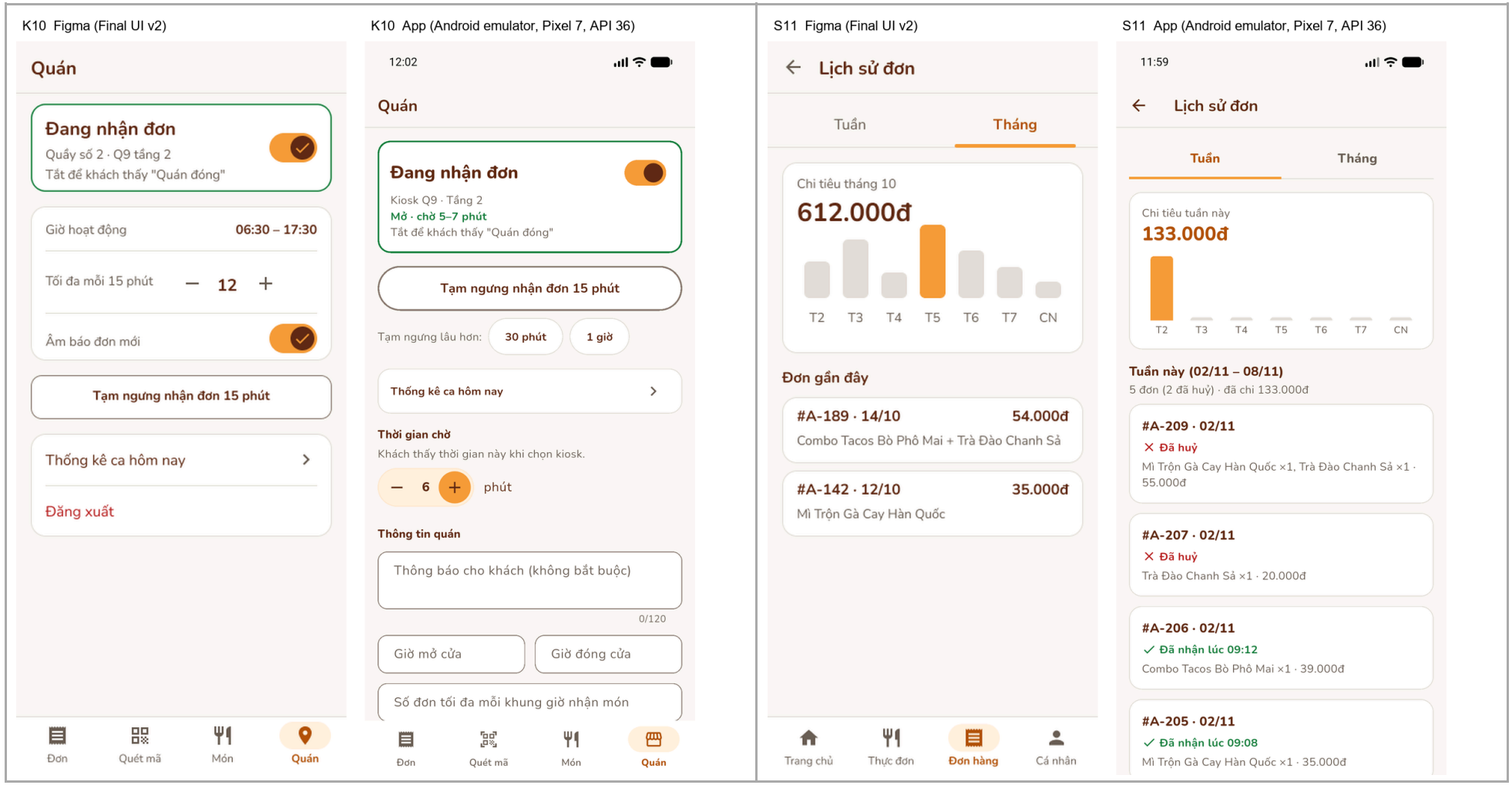
Persona and Lab 2 success criterion

Students who study on campus and have a short break.

- **Goal:** finish an order in **under 35 seconds** with **at most 8 taps**.
- **Customer app** S01–S30: order, pay (simulated), follow, edit, review.
- **Seller app** K01–K15: take orders, menu, shop status, shift statistics.
- 44 of 44 screens built; no placeholder route left.

The `purchase` event measures the criterion (docs/analytics-plan.md).

Figma vs app: seller shop (K10) and history (S11)



Left of each pair: Figma v2. Right: the app (Android emulator, fake data). Differences are recorded as D21 and D25 in docs/design-traceability.md.

Deviations from the design, and why

- **D01/D03** Google Sign-In replaces e-mail + OTP; a signed-out user only sees sign-in (handout).
- **D12/D13** stepper and floating cart button: add, correct and open the cart in reach of the thumb.
- **D18** S25 offers kiosk break slots with a capacity, not a typed timetable (the data model books slots).
- **D23/D24** no photo box, no "Xoá món": no Cloud Storage, and rules forbid deleting menu items.
- **D21–D27** P2 screens add what the data model already had (wait time, notice, hours, timeline, receipt).

33 rows D01–D33 in docs/design-traceability.md section 5. "Figma updated" is still Not yet (T83).

Architecture: MVVM, layers only point down

Screen	draws `ViewState`, forwards taps (`ui/*/widgets`)
ViewModel	state, validation, `Command` (`ui/*/view_models`)
Repository	business data in Firestore (`data/repositories`)
Service	wraps one SDK (`data/services`)
Model	immutable data, pure rules (`domain/models`)

`test/architecture_test.dart` fails on an SDK import in a screen, a colour literal in a screen, or a ViewModel without `SafeChangeNotifier`. Fake data is in `dev/`, outside `lib/`.

T72: 3 pieces of logic moved out of `build()`, 2 kept with a reason (docs/architecture.md section 3).

Design system in code

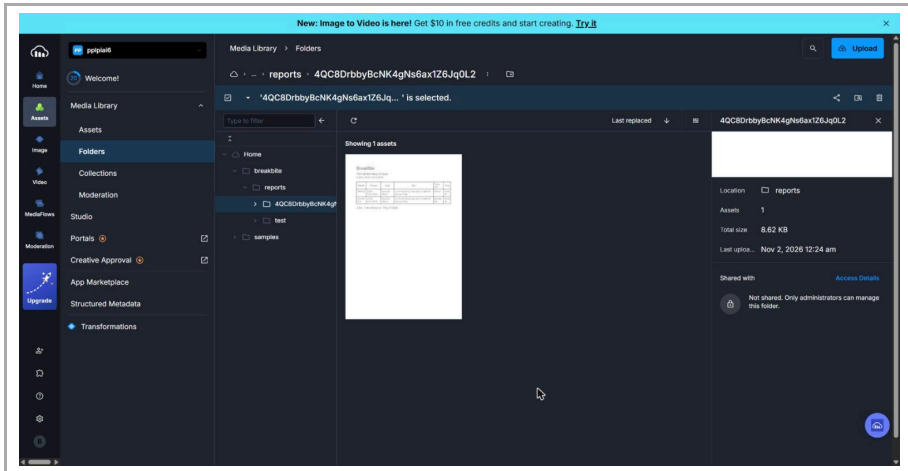
- **Tokens** in `lib/ui/core/theme/`: colors (Figma v2, WCAG contrast tests), dimensions (48 dp tap target), theme.
- **Components** one widget each in `lib/ui/core/widgets/`: button, chip, top bar, stepper, state views.
- **Widget catalog** `dev/widget_catalog.dart`, tested at 360 and 412 dp, text 1.0 and 1.3: no overflow, every target ≥ 48 dp.
- Screens hold no colour or text-style literal (architecture test).

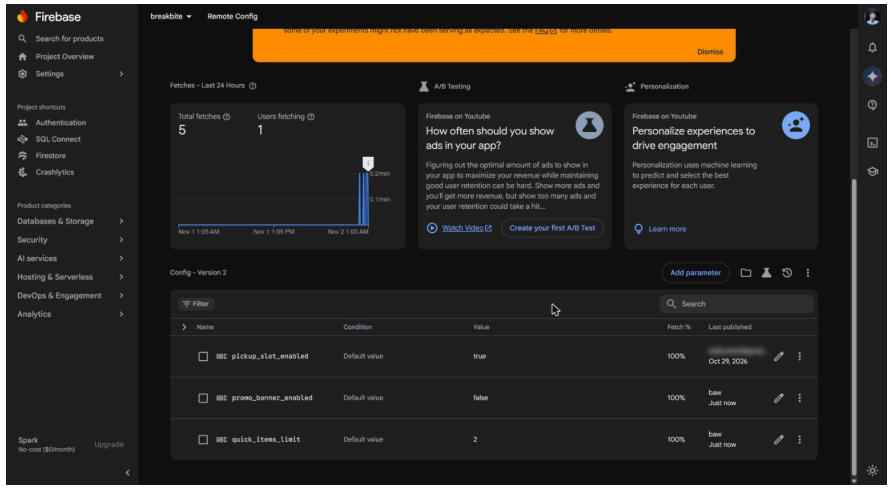
Firebase design: one service, one interface, typed failures

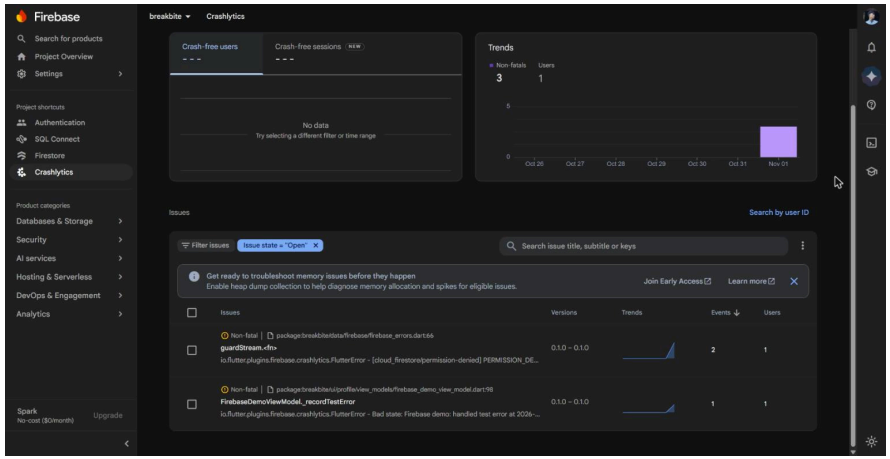
Service	What the app does
Auth	Google Sign-In, role chooses the app; cancelled / offline messages
Firestore	Rules deny by default; orders in transactions; 88 rules tests
Files	Cloudinary behind `StorageService`: PDF with progress, URL, `reports` metadata
FCM	Permission, token, in-app list, tap opens the order
Analytics	11 planned events, all logged, no personal data
Crashlytics	Uncaught + non-fatal with keys `role`, `kiosk_id`
Remote Config	3 parameters, Home changes without a rebuild

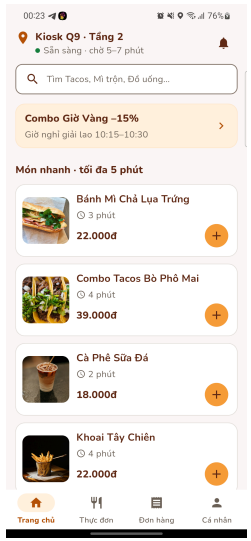
Why Cloudinary: Cloud Storage needs the Blaze plan. Details per service: docs/firebase-design.md.

Firebase evidence (project breakbite-1315b)









Uploaded PDF in the Cloudinary library · Remote Config console · non-fatal error in Crashlytics · signed-in app.

Analytics: 11 events, each answers a question

- **Sign-in/out:** `login`, `logout` · **Find:** `search`, `view_menu_item`
- **Order:** `add_to_cart`, `reorder`, `begin_checkout`, `purchase`, `payment_failed`
- **Seller:** `order_accepted`, `order_delivered` · **Files:** `export_report`
- `purchase` tests the success criterion (≤ 8 taps, < 35 s). `setUserId` after sign-in.

All 11 are logged in code and tested in ViewModels. DebugView screenshots: not collected yet.

Testing: what ran, what did not

Check	Result
Unit, ViewModel, widget, flow tests	749 pass (`flutter test`)
Firestore rules tests	88 pass
Integration vs real rules (emulators, Android emulator)	15 pass
ARTEMIS: 31 cases written (TC-01–TC-31)	**Not run yet** (needs server, model key, device)

Each ARTEMIS test asserts on the value the agent reports; results table comes from run_repeat.py (3 runs each).

ARTEMIS: cases and exploratory session

- 10/11/2026, ~10 minutes, emulator with fake data, text at 200 %, fast taps, odd search, receipt export.
- **5 low-severity problems:** clipped tab label and status line, a misleading empty-search text, contradictory order wording, K10 labels cut.
- **1 fixed** (K10 stacks the time fields, test added); 4 open with an owner.
- 31 ARTEMIS cases written, **0 run**; P5 repeats this session with ARTEMIS on a phone.

Details: docs/test-plan.md section 4.

AI code review: 10 findings, 5 fixed

ID	Type	Decision	Fix
CR-01 order prices from client	security	Accepted, known limitation	—
CR-02 new item overwrites old	bug	Accepted	d6d8113
CR-03 stale incident answered twice	bug	Accepted	d1291a6
CR-04 batch over 500 writes	performance	Accepted	9243bf3
CR-05 older orders button does nothing	bug	Accepted	2980559
CR-06 time ignores seconds	bug	Accepted	ba8dc33
CR-07 – CR-10	mixed	Rejected, each with a reason	—

Tool: Claude Code /code-review. Output: evidence/code-review/. Table and reasons: docs/code-review.md.

Challenges and lessons learned

- **Cloud Storage costs money** → Cloudinary behind the same interface; lecturer's agreement still to attach.
- **Check in the UI is not a rule** → real rules on the emulator found gaps the fakes hid (CR-01, CR-03).
- **Stale copies** → read again inside a transaction.
- **Write the plan first** → traceability and test plan before screens; every late change cost a row and a Figma edit.
- **An AI review is questions, not facts** → 4 of 10 findings rejected after reading the code.

What is still missing (no evidence → not claimed)

- ARTEMIS runs and pass-rate table (P5) · Analytics DebugView · FCM screenshots (P5)
- Real-Firebase run of the seller screens with a staff account (P3 sets the role)
- Receipt upload from S12 in the Cloudinary library (P2)
- Lecturer's written confirmation (T11) · Figma update (T83) · demo video (T82)

Full list with owners: report section 11.

Where to find everything

Looking for	File
Traceability and deviations	`docs/design-traceability.md`
Architecture	`docs/architecture.md`
Firebase design, analytics plan	`docs/firebase-design.md`, `docs/analytics-plan.md`
Test plan, ARTEMIS tests	`docs/test-plan.md`, `e2e_test/`
Code review	`docs/code-review.md`, `evidence/code-review/`
Evidence	`evidence/`
Demo video	link in README.md (not recorded yet)

Slides and report were generated with Claude Code from these files (scripts/build_report.py).